

PFC AI Design Agent

Report Structure & Documentation Agent

Agreed Specification

A complete record of all documentation and report structure decisions
agreed during the planning discussions of June 2026

Status: Planning — No source code changes have been made

Version 1.0 · June 2026 · PFC Engineering Team

This document records every decision made during the documentation planning sessions. It is the single reference point before any implementation begins. All items marked ✓ AGREED are confirmed and locked. Open items are flagged.

Part 1 — Summary of All Agreed Decisions

The following table is a quick-reference summary. Every row is elaborated in the detailed sections that follow.

Decision Area	What Was Agreed	Status
Documentation Agent	Option A — dedicated sub-agent, scope-limited to report generator files only. Triggered per step as designer approves.	✓ AGREED
Report format	Chapter-based (not "Step X" numbering). 6 chapters plus front and back matter.	✓ AGREED
Numbering style	Three-level decimal: X.Y.Z (e.g. 3.4.2)	✓ AGREED
Front matter	Cover page, revision history, executive summary scorecard, table of contents, nomenclature, abbreviations	✓ AGREED
Back matter	Appendix A: BOM, Appendix B: Bench plan, Appendix C: Sensitivity analysis, Appendix D: References	✓ AGREED
Chapter 1 title	Specifications	✓ AGREED
Chapter 2 title	Topology and Control Scheme Selection	✓ AGREED
Chapter 3 title	PFC Inductor Sizing	✓ AGREED
Chapter 3 scope	All selection decisions and calculations: inductance target, ripple, K(D), material, wire, core, turns, fill, first-pass loss/thermal tables.	✓ AGREED
Chapter 4 title	PFC Inductor Performance Analysis	✓ AGREED
Chapter 4 scope	All graphs, waveforms, 3D model, wire fitting view, magnetic performance under all conditions. Corresponds to JS studio output.	✓ AGREED
Chapter 4 tables	Detailed cycle-averaged tables go in Chapter 4. First-pass peak-point tables stay in Chapter 3.	✓ AGREED
JS studio note	No attribution note needed in Chapter 4 for the JS studio.	✓ AGREED
Chapter 5 title	DC Bus Capacitor Selection	✓ AGREED
Chapter 6 title	Control Scheme	✓ AGREED
Structural elements	Five annotation boxes throughout: CONCEPT, THEORY, PITFALL, DECISION, INSIGHT	✓ AGREED
Writing style	Numbered sub-steps (Step 1, Step 2...), every equation followed by numerical substitution, Decision box at end of every calculation	✓ AGREED
Agent role definition	To be defined further as process develops	In progress

Part 2 — Documentation Agent Definition

✓ AGREED Option A Selected — Dedicated Sub-Agent

A dedicated Documentation Agent is created as a Claude Code project skill. It has one job: improving and generating the report content. It never touches calculation logic, frontend components, or API endpoints.

2.1 Agent Identity and Scope

The Documentation Agent is defined by three hard constraints:

- **Can ONLY edit:** generate_report.py, generate_steps13_14.py, generate_step15.py, generate_step16.py, generate_combined_report.py, and Markdown documentation files (README.md, DEVELOPMENT_GUIDE.md, API_REFERENCE.md)
- **Can NEVER edit:** step7_magnetic_calc.py, step15_capacitor.py, step16_control_design.py, main.py, any frontend .tsx/.ts file, client.ts, any test file, any configuration file
- **Can NEVER change:** calculation logic, data structures, API contracts, GUI behaviour, or database schemas

2.2 When the Agent Works

The Documentation Agent is triggered at the moment of designer approval — not before. The workflow is:

Step	Actor	Action	Documentation Agent Role
1	Designer	Completes a design step in the GUI (e.g. selects core in Step 7)	Inactive
2	Designer	Reviews the result on screen, satisfied with the design	Inactive
3	Designer	Clicks Approve — the approved result is stored in App state	Activated
4	Agent	Collects all data for this step from the approved result object	Data collection
5	Agent	Generates the corresponding report chapter/section with full narrative	Content generation
6	Designer	Reviews and optionally adjusts the generated content	Awaiting review
7	Designer	Approves content — it is committed to the report generator	Content locked

2.3 Agent Knowledge Base

The Documentation Agent is pre-loaded with the following knowledge so it never needs to be re-briefed:

- **The five structural elements** — CONCEPT, THEORY, PITFALL, DECISION, INSIGHT — with examples from the four reference documents
- **Three-level numbering rules** — X.Y.Z within each chapter
- **Chapter scope boundaries** — exactly what belongs in each chapter
- **Writing style guide** — sentence-level rules, table rules, figure rules
- **The 21-item improvement checklist** — what to add to each section
- **Reference document patterns** — from EDGE magnetic report, FAN9672 control loop report, capacitor calculation document
- **File locations** — which Python generator file corresponds to which chapter

2.4 Agent Role — To Be Defined Further

INSIGHT Progressive Role Definition

The agent role will be refined as the process develops. The initial scope covers report content generation. Future scope may include documentation audit (checking generated content against the standard), cross-chapter consistency checking, and automatic DECISION box population from calculation result data.

Part 3 — Report Structure

✓ AGREED Chapter-Based Format with Three-Level Numbering

The report uses chapter-based organisation (Chapter 1 through Chapter 6) with three-level decimal numbering (X.Y.Z). The current "Step X" labels are replaced entirely.

3.1 Complete Report Structure Map

FRONT MATTER

- Cover page — document title, project name, engineer, date, revision, status
- Revision history table — what changed in each generated revision
- Executive Summary — one-page scorecard of all key results with PASS/FAIL verdicts
- Table of Contents
- Nomenclature table — every symbol, its meaning, units, and value in this design
- Abbreviations table

Chapter	Title	Core Question	Approx. Pages
1	Specifications	What must this design achieve?	3–5
2	Topology and Control Scheme Selection	What architecture do we use and why?	5–7
3	PFC Inductor Sizing	What inductance and winding do we need?	10–14
4	PFC Inductor Performance Analysis	How does the chosen inductor behave across all conditions?	12–18
5	DC Bus Capacitor Selection	How do we size and qualify the capacitor bank?	8–12
6	Control Scheme	How do we close the loops stably across all conditions?	12–16

BACK MATTER

- Appendix A — Full component bill of materials (BOM)
- Appendix B — Bench measurement and validation plan
- Appendix C — Sensitivity and tolerance analysis (consolidated)
- Appendix D — Reference standards and application notes cited

Part 4 — Chapter-by-Chapter Detailed Outline

Each chapter heading below shows the complete three-level section structure and the key content agreed for that chapter.

Chapter 1

Specifications

What must this design achieve?

Scope

Chapter 1 records all fixed input requirements — electrical, mechanical, regulatory — that every subsequent chapter must satisfy. Nothing is calculated here. Every number in this chapter comes directly from the customer specification or the applicable standard.

- **1.1 Project Identification**

- 1.1.1 Project name, engineer, date, tool version
- 1.1.2 Design scope and deliverable description

- **1.2 Input / Output Electrical Requirements**

- 1.2.1 Input voltage range ($V_{in,min}$ to $V_{in,max}$)
- 1.2.2 Output voltage V_{out} and tolerance
- 1.2.3 Output power — low line and high line
- 1.2.4 Efficiency targets per operating point
- 1.2.5 Power factor requirement

- **1.3 Compliance and Standards**

- 1.3.1 IEC 61000-3-2 harmonic limits — class and limit table
- 1.3.2 SEMI F47 voltage sag ride-through — sag profile and hold-up requirement
- 1.3.3 Any additional applicable standards (medical, telecom, etc.)

- **1.4 Operating Points Matrix**

- 1.4.1 The 9-point operating matrix — V_{in} , P_{out} , η , PF for each point
- 1.4.2 Low-line family (90–132 Vac, 1700 W) and high-line family (180–264 Vac, 3600 W)
- 1.4.3 Note on the 132–180 Vac gap — no interpolation performed

- **1.5 Thermal and Mechanical Constraints**

- 1.5.1 Ambient temperature range

- 1.5.2 Maximum component temperatures
- 1.5.3 Form factor and height constraints
- **1.6 Design Targets Summary**
 - 1.6.1 One-table summary of all inputs — the single reference for all chapters

Chapter 2

Topology and Control Scheme Selection

What architecture do we use and why?

Scope

Chapter 2 documents all architecture decisions made in the Mode A gate process — topology, operating mode, controller IC, channel count, switching frequency. Each decision is documented with its rationale, the alternatives considered, and the margin or benefit it provides.

- **2.1 Topology Selection**
 - 2.1.1 Candidate topologies evaluated (single-phase boost, interleaved boost, bridgeless, totem-pole)
 - 2.1.2 Scoring criteria and weighting
 - 2.1.3 Selected topology — interleaved boost — with rationale
- **2.2 Operating Mode**
 - 2.2.1 CCM vs CrCM — trade-offs at this power level
 - 2.2.2 Selected mode: CCM — rationale and operating boundary
- **2.3 Controller IC Selection**
 - 2.3.1 Analog vs digital control
 - 2.3.2 Selected controller: FAN9672 — key features used
 - 2.3.3 Control architecture overview — ACM two-loop block diagram
- **2.4 Channel Count**
 - 2.4.1 1-phase vs 2-phase vs 3-phase — ripple cancellation and complexity
 - 2.4.2 Selected: 2-phase interleaved — ripple reduction and EMI benefit
- **2.5 Switching Frequency**
 - 2.5.1 The four-way frequency trade-off: core loss, copper loss, gate drive loss, EMI filter size
 - 2.5.2 Selected: 70 kHz per phase — rationale and sensitivity
- **2.6 Architecture Summary**
 - 2.6.1 Block diagram of the selected architecture
 - 2.6.2 Key design constants table — VRAMP, GMI, GMV, KRLPK, KRM

Chapter 3

PFC Inductor Sizing

What inductance and winding do we need?

Scope — Agreed

✓ AGREED Chapter 3 = Selection and Calculation Decisions

Chapter 3 contains all the design decisions and calculations that lead to "we choose this core, this wire, this turns count." It ends with the approved design declared. Chapter 3 includes first-pass performance tables (peak-point estimates) that were used to make the selection decision.

• 3.1 Inductor Design Requirements

- 3.1.1 Target inductance derivation — governing equation with full derivation
- 3.1.2 Per-phase operating currents — I_{pk} , I_{rms} at all 9 operating points
- 3.1.3 DC bias current at crest — $I_{avg@crest}$ for each V_{in}
- 3.1.4 Current density target — J_{target} and its physical basis

• 3.2 Ripple and Interleaving Analysis

- 3.2.1 Per-phase inductor ripple $\Delta I_L(t)$ — governing equations
- 3.2.2 Interleaving cancellation factor $K(D)$ — derivation and physical meaning
- 3.2.3 $K(D)$ at each operating point — table
- 3.2.4 Worst-case line angle for maximum per-phase ripple
- 3.2.5 Input ripple after cancellation — verification against target

• 3.3 Core Material Selection

- 3.3.1 Material type: powder vs ferrite — DC bias characteristic comparison
- 3.3.2 Soft saturation vs hard saturation — physical explanation
- 3.3.3 Selected material family and grade
- 3.3.4 Material model — Steinmetz constants validated against datasheet
- 3.3.5 Permeability vs DC bias model — validated against catalog spec points

• 3.4 Core Geometry Selection

- 3.4.1 Sizing engine inputs — L_{target} , I_{peak} , J_{target} , f_{sw} , $T_{operating}$
- 3.4.2 Top candidates table — all metrics side by side
- 3.4.3 Candidate comparison — loss, fill factor, ΔT , geometry, stacking
- 3.4.4 Stacking note — window area does NOT multiply with stack count
- 3.4.5 Selected core — part number, dimensions, AL , A_e , L_e , V_e , W_a
- 3.4.6 Selection rationale and design margins

• 3.5 Winding Design

- 3.5.1 Wire type selection — Litz vs solid vs TIW and why
- 3.5.2 Skin depth at f_{sw} — strand diameter requirement
- 3.5.3 Strand count and bundle OD
- 3.5.4 Number of turns N — from AL sizing
- 3.5.5 Window fill factor FF_{cu} — bare copper fill
- 3.5.6 Insulated bundle fill — practical winding limit check
- 3.5.7 Layer count screening

- **3.6 First-Pass Loss and Thermal Estimates**

- 3.6.1 Copper loss estimate — $I_{rms}^2 \times DCR$ at 100°C
- 3.6.2 Core loss estimate — peak-point Steinmetz (used for selection only)
- 3.6.3 Total loss estimate and thermal check — $\Delta T < \text{budget?}$
- 3.6.4 Loss vs input voltage table — all 9 points, min/nom/max AL (first-pass)

- **3.7 Sizing Summary**

- 3.7.1 Approved design parameters — complete table
- 3.7.2 Design margins overview — fill, saturation, temperature
- 3.7.3 Correction log — any corrections applied from previous iteration

Chapter 4

PFC Inductor Performance Analysis

How does the chosen inductor behave across all conditions?

Scope — Agreed

✓ AGREED Chapter 4 = Waveforms, Graphs, 3D Model, Detailed Performance

Chapter 4 contains everything the JS Review studio produces. When the designer approves the design on the Review page, Chapter 4 is generated from that data. All waveform plots, 3D models, window build views, and sweep plots become permanent figures in this chapter. Detailed cycle-averaged tables (more accurate than the first-pass tables in Chapter 3) are also here.

INSIGHT Chapter 4 = Engineering Proof

Chapter 3 is the engineering judgement — the decision. Chapter 4 is the engineering proof — the evidence. A reviewer who trusts the Chapter 3 decision reads Chapter 4 to verify it. A reviewer who questions the Chapter 3 decision reads Chapter 4 to challenge it.

• 4.1 Physical Model and Assembly

- 4.1.1 3D rendered model — core and winding assembly
- 4.1.2 Window build — layer-by-layer wire fit cross-section view
- 4.1.3 Assembled dimensions — OD, height, bore hole remaining

• 4.2 Inductance Performance

- 4.2.1 No-load inductance L_0 — AL tolerance band (min/nom/max)
- 4.2.2 L vs DC bias — retention $k(H)$ at all 9 operating points
- 4.2.3 L_{full} vs input voltage — with AL tolerance band
- 4.2.4 L variation: no-load to full-load — range and percentage

• 4.3 Flux Density Analysis

- 4.3.1 Governing equations — ΔB , B_{dc} , $B_{ac, pk}$, B_{max}
- 4.3.2 $B_{ac, pk}(t)$ waveforms over half line cycle — all input voltages
- 4.3.3 $B_{dc}(t)$ and $B_{max}(t)$ — DC-bias excursion with ripple band
- 4.3.4 Flux density summary table — all 9 operating points
- 4.3.5 Saturation margin — B_{max} vs B_{sat} , L vs I curve with operating point

• 4.4 Magnetising Field $H(t)$

- 4.4.1 $H(t)$ waveforms over half line cycle — all input voltages
- 4.4.2 Peak H vs permeability knee — operating point on $k(H)$ curve
- 4.4.3 H summary table — peak values at all 9 operating points

• 4.5 Core Loss — Time-Domain iGSE

- 4.5.1 iGSE model — governing equations and F(D) correction derivation
- 4.5.2 F(D) values at all 9 operating corners
- 4.5.3 $P_{core}(t)$ waveforms — all input voltages (cycle-averaged)
- 4.5.4 The double-hump signature at high line — explained
- 4.5.5 Cycle-averaged vs peak-point core loss — comparison table
- 4.5.6 Core loss vs input voltage — typical and max-anchored

• 4.6 Copper Loss Analysis

- 4.6.1 $P_{cu}(t)$ waveforms — DC + AC/HF decomposition
- 4.6.2 Copper loss at 25°C, 80°C, and 100°C — all 9 points
- 4.6.3 HF (skin and proximity) copper loss component
- 4.6.4 DCR temperature coefficient — $R_{dc}(T)$ formula and table

• 4.7 Total Loss and Thermal Performance

- 4.7.1 $P_{total}(t) = P_{core}(t) + P_{cu}(t)$ — instantaneous waveforms
- 4.7.2 Total loss vs input voltage — cycle-averaged, all 9 points
- 4.7.3 Loss uncertainty band — $\pm 5\text{--}20\%$ and its sources
- 4.7.4 Surface area calculation — SA formula and geometric derivation
- 4.7.5 Temperature rise ΔT vs input voltage
- 4.7.6 Iterated thermal solution — converged winding temperature
- 4.7.7 Thermal margin vs budget

• 4.8 Operating Boundary Verification

- 4.8.1 CCM/DCM boundary check — ΔI_{pp} vs $2 \times I_{avg}$ at all operating points
- 4.8.2 CCM validity across the half line cycle

• 4.9 Design Validation

- 4.9.1 Sensitivity and tolerance analysis — AL $\pm 8\%$, temperature, material variation
- 4.9.2 Material model validity — μ vs frequency, Steinmetz range
- 4.9.3 Design audit PASS/FAIL checklist — each criterion with source cited
- 4.9.4 Bench measurement and validation plan

Chapter 5

DC Bus Capacitor Selection

How do we size and qualify the capacitor bank?

Scope

Chapter 5 sizes the bulk output capacitor, selects a real part from the database, verifies performance, and qualifies the bank for the required service life.

• 5.1 Capacitor Sizing Requirements

- 5.1.1 Hold-up time energy balance — full derivation from first principles
- 5.1.2 $C_{\text{hold-up}}$ calculation — worst-case operating point
- 5.1.3 Ripple voltage sizing — C_{ripple} calculation
- 5.1.4 Governing capacitance — which requirement is larger and why

• 5.2 Bank Configuration Decision

- 5.2.1 Single cap vs parallel bank — ripple current sharing rationale
- 5.2.2 Selected bank: quantity × value
- 5.2.3 Voltage rating and derating

• 5.3 Part Selection

- 5.3.1 Filtering criteria — voltage, lifetime, ESR, ripple rating
- 5.3.2 Selected part — all datasheet parameters tabulated

• 5.4 Ripple Current and Voltage Verification

- 5.4.1 LF ripple current ($2 \times f_{\text{line}}$) derivation
- 5.4.2 HF ripple current (switching) derivation
- 5.4.3 Total I_{rms} — LF and HF add in quadrature
- 5.4.4 Per-capacitor share — utilisation vs rating
- 5.4.5 Ripple voltage verification against specification

• 5.5 Capacitor Lifetime Analysis

- 5.5.1 Physical basis — why electrolytic capacitors wear out (Arrhenius)
- 5.5.2 Method 1 — Simplified Arrhenius from datasheet ESR
- 5.5.3 Method 2 — Simplified Arrhenius from $\tan\delta$ -derived ESR
- 5.5.4 Method 3 — Full manufacturer model $L = L_0 \cdot f(T) \cdot f(I) \cdot f(V)$
- 5.5.5 Three-method comparison — confidence band interpretation
- 5.5.6 Pass/fail against 15-year design target

• 5.6 Capacitor Bank Summary

- 5.6.1 Approved bank configuration — all parameters
- 5.6.2 Design margins — ripple current, voltage, lifetime

Chapter 6

Control Scheme

How do we close the loops stably across all conditions?

Scope

Chapter 6 designs the average-current-mode control loops for the FAN9672 controller, verifies stability at all 9 operating points, and provides a complete control network bill of materials.

• 6.1 Control Architecture

- 6.1.1 Average current-mode control — what it does and why it is used
- 6.1.2 The two-loop structure — inner current loop and outer voltage loop
- 6.1.3 The GMOD block — role in maintaining loop gain across the line cycle
- 6.1.4 How to read a Bode plot — for readers unfamiliar with frequency-domain analysis
- 6.1.5 Phase margin and gain margin — physical meaning

• 6.2 Plant Analysis

- 6.2.1 Right-half-plane zero — physical origin and bandwidth limitation
- 6.2.2 RHP zero — current path vs voltage plant (different inductance values)
- 6.2.3 LC double pole frequency f_0
- 6.2.4 ESR zero frequency f_{esr}
- 6.2.5 RHP zero at all 9 operating points — table
- 6.2.6 Recommended bandwidth targets — derivation and rationale

• 6.3 FAN9672 Pin Configuration

- 6.3.1 Pin-by-pin table — every pin with its function and external component value
- 6.3.2 Operating envelope parameters — R_{CS} , K_{max} , V_{EA} window
- 6.3.3 G_{MOD} verification — three-path self-audit
- 6.3.4 Supporting components — R_{GC} , R_{LS} , R_{RI} , R_{RLPK} sizing

• 6.4 Current Loop Design

- 6.4.1 Type II compensator topology — why Type II is sufficient for the current loop
- 6.4.2 Crossover frequency target and rationale
- 6.4.3 Component sizing — R_{IC} , $C1_{\text{IC}}$, $C2_{\text{IC}}$ with full derivation
- 6.4.4 Achieved phase and gain margins at sizing corner
- 6.4.5 Current loop Bode plot — LL and HL corners

• 6.5 Voltage Loop Design

- 6.5.1 Type II vs Type III OTA — decision criterion and selected type
- 6.5.2 SLVA662 Method B — derivation summary
- 6.5.3 Component sizing — $R2$, $C1$, $C3$ (Type II) or $R2$, $R3$, $C1$, $C2$, $C3$ (Type III)

- 6.5.4 120 Hz rejection requirement — physical basis and 20 dB target
- 6.5.5 Voltage loop Bode plot — LL and HL corners

- **6.6 Stability Scorecard**

- 6.6.1 Why all 9 operating points must be checked
- 6.6.2 Pass criteria — $PM \geq 45^\circ$ (current), $PM \geq 60^\circ$ (voltage), 120 Hz rej ≥ 20 dB
- 6.6.3 Full 9-point scorecard table with PASS/FAIL

- **6.7 Soft-Start and Protection**

- 6.7.1 Soft-start capacitor C_{SS} — sizing and ramp time
- 6.7.2 ILIMIT and ILIMIT2 — current clamp sizing
- 6.7.3 BIBO — brown-in/brown-out threshold calculation

- **6.8 Control Network BOM**

- 6.8.1 All compensator components — reference, value, function
- 6.8.2 Feedback divider components
- 6.8.3 Soft-start and protection components

Part 5 — Five Structural Elements

✓ AGREED Five Annotation Boxes Used Throughout All Chapters

Every chapter uses the same five annotation boxes. They appear in a consistent visual style so any reader immediately recognises their meaning.

Box Type	Tag	When to Use	Content Rule
CONCEPT	Blue left border	Start of every calculation step	2–4 sentences. WHY this calculation exists. No equations. Readable by anyone who knows what a switching converter is.
THEORY	Green left border	When a governing equation needs physical derivation	The mathematical basis — where the equation comes from. Reference the standard or app note if applicable.
PITFALL	Red left border	Where a common mistake exists	One specific, named mistake. State what happens if you make it (e.g. "+83% error in core loss"). Not generic warnings.
DECISION	Navy left border	End of every calculation step	Selected value, standard rounding, margin provided, rationale. This is the permanent design audit trail.
INSIGHT	Amber left border	Non-obvious observations	Engineering knowledge that makes the reader better. Not a calculation — an observation from the results.

5.1 Numbered Sub-Step Format

Every calculation follows this exact sequence inside each section:

- **Opening sentence** — one line stating what this section computes
- **CONCEPT box** — why this calculation exists
- **THEORY box** (if needed) — derivation of the governing equation
- **Governing equation** — symbolic form with all variables in the nomenclature table
- **Step 1, Step 2, Step 3...** — one numerical operation per step, result shown
- **PITFALL box** (if applicable) — the most common error at this step
- **DECISION box** — what was chosen and why
- **INSIGHT box** (if applicable) — non-obvious observation about the result

5.2 Table Design Standard

- Every table has an **assessment or verdict column** when the result is a criteria check
- **Worst-case row highlighted** in amber or bold
- **Min / nominal / max columns** for any value with tolerance (AL $\pm 8\%$, etc.)
- **Units in column header**, not repeated in every cell

- Every table **referenced by number** in the narrative before it appears

5.3 Figure Standard

- Every figure has a **number and full descriptive caption** — not just a title
- Every figure **referenced by number in the narrative** before it appears
- **Worst-case curve labelled** on every multi-curve plot
- **Limit / target line shown** as horizontal dashed on every margin plot

Document Status: Agreed decisions locked. Agent role and further chapter detail to be refined as implementation progresses. No source code changes have been made as of this document.

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